Zhantong Qiu

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Introduction

Computer architecture researcher focusing on computer system performance evaluation, simulation methodology, and hardware-software co-design for emerging applications. I build novel frameworks for fast, accurate, and portable computer system performance evaluation using sampled simulation. I tackle robotic applications on tiny robots and HPC workloads in datacenters through hardware-software co-design. I am a strong programmer and active contributor to production open-source simulators (e.g., gem5). I seek challenges in constrained systems and solve them with solutions that span multiple layers of the computer system stack.

Education

Master of Science in Computer Science — Fall 2023 – Spring 2026 (expected)
University of California, Davis *Davis, California*

Visiting Student in Electrical and Computer Engineering — Fall 2025 – Spring 2026
Cornell University *Ithaca, New York*

Bachelor of Science in Computer Science and Engineering — Fall 2020 – Spring 2023
University of California, Davis *Davis, California*

Publications

- **Nugget: Portable Program Snippets**[5] (HPCA 2026)
  Zhantong Qiu, Mahyar Samani, Jason Lowe-Power.
  Introduced an LLVM IR–based sampling framework that converts long-running workloads into portable, binary/ISA-independent "nuggets," enabling hardware-based interval analysis and reducing sampling overhead by up to over 500× vs. functional simulation; used to develop sampling methods and diagnose simulator modeling errors in gem5.

- **Accelerating the Simulation of Parallel Workloads using Loop-Bounded Checkpoints** (TACO 2025)
  Alen Sabu, **Zhantong Qiu**, Harish Patil, Changxi Liu, Wim Heirman, Jason Lowe-Power, Trevor E. Carlson.
  Paper extension of *LoopPoint*, a synchronization-agnostic loop-based sampling method with loop-bounded checkpoints for fast, accurate multi-threaded simulation, delivering up to 801× speedup on SPEC CPU2017 with ∼2.3% average runtime error; released a *gem5* module for full-system simulation and introduced *ROIperf* to validate representative regions directly on silicon.

Talks and Tutorials

- **HPCA 2026** – *Nugget: Portable Program Snippets* (Main Conference, Sydney, Australia).
- **CGO 2026** – Nugget with LLVM infrastructure (LLVM–CGO 2026 Workshop, Sydney, Australia).
- **gem5 Bootcamp 2024** – Week-long hands-on training on the gem5 simulator (UC Davis, CA).
- **HPCA 2023** – LoopPoint using Sniper and gem5 (LoopPoint Tutorial, Montreal, Canada).
- **ISCA 2023** – Technique talk on full-system sampling support in gem5 (gem5 Workshop, Orlando, FL).

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[5] https://arxiv.org/abs/2509.02873

Research Experience

Graduate Researcher June 2023 – Present

DArchR Lab, UC Davis

Advisor: Jason Lowe-Power

- **Nugget (LLVM IR sampling)**
  - Designed and implemented a cross-platform, architecture-independent sampling framework at the LLVM IR level; supports rapid interval analysis, native execution for validation, and cross-ISA simulation.
  - Evaluated Nugget on real hardware platforms with diverse performance characteristics (e.g., Ampere Altra) and in the gem5 simulator to validate fidelity and portability.
  - Analyzed a diverse set of workloads (SPEC CPU2017, NPB, LSMS) to demonstrate robustness across CPU, HPC, and scientific applications.
- **Accelerating the Simulation of Parallel Workloads using Loop-Bounded Checkpoints**
  - Co-led methodology and implementation to validate LoopPoint samples on gem5.
- **SPEAR: Scalable Power and Energy Analysis and Performance Tools for Frontier**
  - Collaborating with researchers at Oak Ridge National Laboratory and the University of Wisconsin–Madison to apply sampling techniques (e.g., Nugget) to bound significant execution regions for GPU+CPU high-performance applications on Frontier, a state-of-the-art heterogeneous supercomputer.
  - Characterizing GPU+CPU high-performance applications to develop simulation methodologies that accelerate large-scale system simulation (e.g., Frontier).
- **gem5 contributions**
  - Ongoing upstream contributions since 2022 to full-system sampling support and related simulation features in gem5; author of 50 commits[6] (public history).

Visiting Student August 2025 – Present

Computer Systems Laboratory, Cornell University

Advisor: Christopher Batten

- **Agile Robotic Hardware-Software Co-Design**
  - Developing a closed-loop, multi-robot evaluation framework integrating robotics simulators and architecture simulators; enables parallel per-robot simulation and decouples from a specific ISA/simulator to broaden studies (e.g., ISA extensions).
- **Accurate STM32G4 Board in gem5**
  - Building an accurate STM32G4 MCU model in gem5 (Arm M-profile). Implemented the ART (Adaptive Real-Time) accelerator[7] to improve Flash access latency; planning to upstream core M-profile components (e.g., NVIC).

Undergraduate Researcher June 2022 – June 2023

DArchR Lab, UC Davis

Advisor: Jason Lowe-Power

- **SimPoint**
  - Implemented SimPoint support in the gem5 stdlib and ran SPEC CPU2006 experiments using SimPoint.
- **LoopPoint**
  - Implemented LoopPoint support in system-emulation and full-system modes in gem5; added a generalized PC-execution counter to track occurrences of specific PCs.

[6] https://github.com/gem5/gem5/commits?author=studyztp

[7] https://github.com/studyztp/gem5/tree/studyztp/ART-cache

Course Projects

CXL Shared Memory Filesystem Optimization[8]
Investigated application needs in a CXL shared-memory system and improved the FAMFS framework with more efficient allocation/deallocation and zero-copy operations to avoid redundant copies.
Limitations of Disaggregated Memory and Innovations[9]
Analyzed limitations of disaggregated memory and proposed a hardware-software co-designed page management approach.
CUDA Microbenchmarks[10]
Developed configurable microbenchmarks to measure shared-memory latency and memory-scaling behavior on GPUs.
RISC-V Operating System
Built a simple RISC-V OS on the UC Davis RISC-V Console Simulator for a course project.

Teaching Experience

WQ 2024 ECS 154B: Computer Architecture Teaching Assistant
Led weekly discussion sections and office hours; created assignment material on Chisel-based CPU model (DINO CPU).

Organizations and Service

Founder of Computer Systems Seminar at UC Davis[11]
Organized weekly speaker series with 12+ talks from academia and industry (still on-going).

HPCA 2026 Artifact Evaluation Reviewer
Helped reviewing the artifact of the accepted paper.

Skills

Programming Languages: C/C++, Python, Bash, CUDA, assembly, Latex
Hardware Description Languages: Chisel
System Evaluation Tools: gem5, QEMU
Profiling & Instrumentation: LLVM passes, DynamoRIO, Valgrind, Linux perf, PAPI
Other Linux Related Tools: cpuset, CRIU, cgroups, Docker
Compilers: GCC/G++, GFortran, Clang/LLVM
Languages: English, Cantonese, Mandarin

[8] https://github.com/orgs/ECS-289D/repositories
[9] https://github.com/studyztp/my-course-paper/blob/main/Limitations_of_Disaggregated_Memory_and_Innovations.pdf
[10] https://github.com/studyztp/CUDA_microbenchmarks.git
[11] https://arch.cs.ucdavis.edu/events/computer-system-seminar